

**Cervical air sacs**  
A pair of cervical (neck) air sacs connect to the neck vertebrae.

**Clavicular air sac**  
Extensions of the clavicular air sac run through the humerus (upper wing) bones.

**Trachea**  
The trachea (windpipe) is supported by rings of cartilage to keep it open.

**Clavicular air sac**  
A single clavicular (collar) air sac sits at the base of the windpipe.

**Anterior thoracic air sacs**  
A pair of anterior thoracic (chest) air sacs are at the front of the lungs.

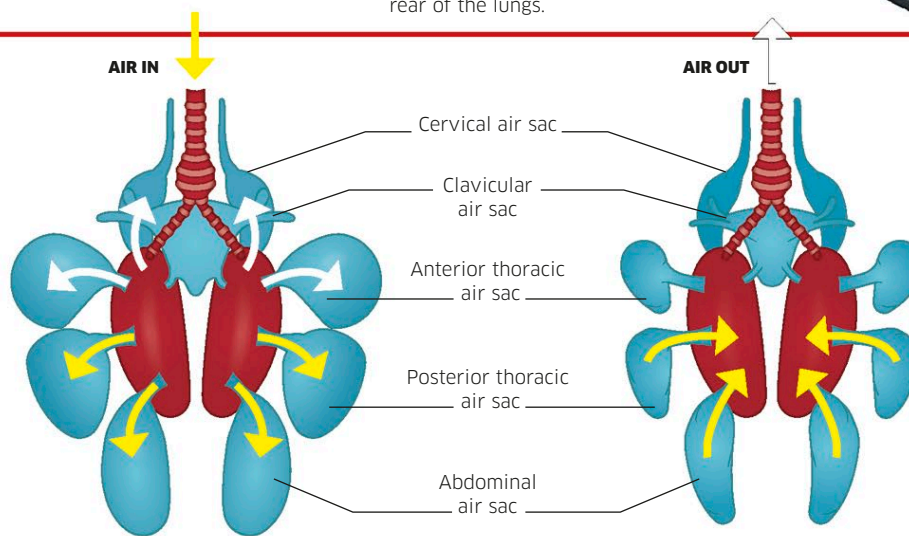
**Posterior thoracic air sacs**  
A pair of posterior thoracic (chest) air sacs are at the rear of the lungs.

### Pigeon's lungs and air sacs

Most birds, like this pigeon, have nine air sacs to pump oxygen-rich air into the lungs. The air sacs in the chest surround the lungs. Others are in the belly and around the windpipe.

### Bird breathing

Humans and other mammals pump air in and out of their lungs using a flat sheet of muscle below the chest called the diaphragm. Birds use air sacs instead to move fresh air through their bodies when breathing in and breathing out. Air circulates one way from the rear air sacs, through the lungs to the front air sacs. The air sacs—which account for 20 percent of the body volume—also help to stop the bird overheating, and make swimming birds buoyant.



#### Inhalation

When a bird breathes in, the air entering the body first fills air sacs at the rear (yellow). At the same time, the air that was already in the lungs moves out to inflate the front air sacs (white).

#### Exhalation

When breathing out, all the air sacs work like bellows to pump air as they deflate. The rear air sacs fill the lungs with air, while the front air sacs push air back out through the mouth.